Einstein Was Doing Machine Learning

The Photoelectric Effect as Statistical Inference

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The Central Claim

Thesis

When Einstein proposed the light-quantum hypothesis in 1905 to explain the photoelectric effect, he was—in the language we now use—doing **machine learning** and **statistical inference**.

Three acts:

1. **Theory** (1905): Einstein proposes the model that defines the hypothesis class
2. **Data quality** (1887–1916): experimenters spend 29 years collecting clean training data
3. **Estimation** (1916): Millikan runs the regression and recovers Planck's constant to 0.5%

The episode illuminates *both* the power of theory-guided learning *and* the limits of purely data-driven AI for scientific discovery.

Background: The Photoelectric Effect

The phenomenon (Hertz, 1887; Hallwachs, 1888):

- Shine light on a metal surface
- Electrons are ejected
- Apply a *stopping potential* V_0 to just halt the fastest electrons

The puzzle (Lenard, 1902):

- V_0 rises with light *frequency* ν
- V_0 is **independent** of light *intensity* I
- Classical wave theory predicted the opposite

Classical Prediction (Wrong)

Wave amplitude $A \propto \sqrt{I}$ delivers energy

$\Rightarrow$ V_0 should rise with I

$\Rightarrow$ V_0 should be independent of ν

What Lenard Found

Exactly the opposite!

Intensity I: **irrelevant**

Frequency ν: **everything**

Lenard's Finding as Feature Engineering

A modern ML framing of Lenard (1902):

Lenard ran an ablation experiment. He measured V_0 as a function of *both* ν and I.

Feature	Effect on V_0	Importance
Frequency ν	Strong $\uparrow$	**High**
Intensity I	None	**Zero**
Wavelength λ	(same as ν)	High

Conclusion: drop I from the feature set; keep ν.

ML Translation

Input features: ν, I, λ, ...

Label: stopping potential V_0

Lenard's result: feature-importance analysis eliminates I; selects ν as the *sole relevant input*

"Lenard ... established **two qualitative facts**: V_0 is independent of intensity; V_0 rises with frequency."

— Millikan (1916)

Einstein's 1905 Insight: Model Architecture

The quantum hypothesis:

Each light quantum carries energy $E = h\nu$.

A single quantum is absorbed by a single electron.

If $h\nu > W$ (work function), the electron escapes.

Energy conservation gives:

$$\underbrace{eV_0}_{\text{max KE}} = \underbrace{h\nu}_{\text{photon energy}} - \underbrace{W}_{\text{work function}}$$

The Prediction (1905)

$$\boxed{V_0 = \frac{h}{e}\nu - \frac{W}{e}}$$

Linear in frequency.

Slope $= h/e$ is **universal** across all metals.

Intercept $= -W/e$ is metal-specific.

Written before a single reliable V_0-vs-ν measurement existed!

The Inductive Bias

The quantum hypothesis restricts the hypothesis class from *all functions* of $(\nu, I, \lambda, \ldots)$ to a **single affine function in ν alone**, with a specific universal slope.

The Regression Equation

Adding measurement noise and a nuisance parameter:

$$V_{0i} = \underbrace{\alpha}_{\text{intercept}} + \underbrace{\beta}_{h/e}\,\nu_i + \underbrace{\varepsilon_i}_{\text{noise}}, \qquad \alpha = -\frac{W}{e} + \underbrace{v_c}_{\text{contact EMF}}$$

Parameters of Interest

$\beta = h/e$	slope $\Rightarrow$ Planck's h
α	intercept (metal-specific)

Nuisance Parameter

v_c: contact EMF between emitter and collector. Can be as large as 3 V—comparable to V_0 itself!

Shifts intercept; **does not bias slope** if

The Supervised Learning Problem

Training data: $\{(\nu_i, V_{0i})\}_{i=1}^{n}$

Task: estimate $\beta = h/e$

Loss: squared error

Hypothesis class: affine functions of ν

Inductive bias: $\beta > 0$; universal across metals

29 Years of Collecting Training Data (1887–1916)

Experimenter	Year	Data quality	Problem
Hertz / Hallwachs	1887–88	Qualitative only	No labeled (ν, V_0) pairs
J. J. Thomson	1899	Identifies electrons	No V_0-vs-ν measurement
Lenard	1902	Qualitative labels	No reliable numerical V_0
Ladenburg	1907	60 nm range	Linear $\approx$ square-root
Hughes	1912	3 UV lines	Contact EMF uncontrolled; 65% scatter in h
Richardson & Compton	1912	8 metals	Same problems; 60% scatter
Millikan	**1916**	**Clean surfaces, wide $\Delta\nu$, controlled v_c**	**h to 0.5%**

Data Quality is the Binding Constraint

The regression is trivial once you have clean data. The 29-year delay reflects *data engineering*, not statistical sophistication.

Millikan's Experimental Design (1916)

Key data-quality innovations ("a machine shop in vacuo"):

- ✓ **Fresh surfaces:** electromagnetic knife shaved the alkali-metal cylinder in vacuum—no oxidation, no work-function contamination
- ✓ **Spectrally pure lines:** mercury arc with filters to exclude UV companions—eliminates stray-light bias on V_0
- ✓ **Wide frequency range:** 5461 Å to 2535 Å ($\Delta\nu = 6.3 \times 10^{14}$ Hz, nearly a factor of two)
- ✓ **Controlled nuisance:** Kelvin double-balance measured v_c continuously; verified stable to ± 0.01 V over 9 hours

Why the Wide Frequency Range Matters (Cramér-Rao)

$$\mathrm{SE}(\hat{\beta}) = \frac{\sigma}{\sqrt{S_{\nu\nu}}}, \quad S_{\nu\nu} = \sum_i (\nu_i - \bar{\nu})^2$$

Wider $\Delta\nu \Rightarrow$ larger $S_{\nu\nu} \Rightarrow$ smaller SE. Hughes's 60 nm range gives a standard error **2.2**× **larger** than Millikan's full range.

Millikan's Training Data

λ (Å)	ν (10^{14} Hz)	V_0^{Na} (V)	V_0^{Li} (V)
5461.0	5.490	0.454	—
4339.4	6.908	1.039	0.499
4046.8	7.409	1.246	0.706
3650.2	8.213	1.578	1.037
3125.5	9.591	2.147	1.606
2534.7	11.827	3.030	2.529

True stopping potential $= V_{0,\mathrm{obs}} + v_c$ (Na: $v_c = 2.51\,\mathrm{V}$; Li: $v_c = 1.52\,\mathrm{V}$)

- 6 labeled training examples per metal
- Two metals (sodium and lithium) available
- Points span nearly a **factor of two** in frequency

OLS Regression: Estimating Planck's Constant

Primary regression: sodium, five points (omitting 2535 Å scattered-light correction)

Standard OLS formula:

$$\hat{\beta} = \frac{S_{\nu V}}{S_{\nu\nu}} = \frac{3.830}{9.277} = \mathbf{4.129 \times 10^{-15}}\ \mathrm{V{\cdot}s}$$

$$\hat{\alpha} = \bar{V}_0 - \hat{\beta}\,\bar{\nu} = -1.813\,\mathrm{V}$$

Statistic	Value
R^2	≈ 0.99999
Residuals	$\leq 2\,\mathrm{mV}$
SE($\hat{\beta}$)	4.6×10^{-18} V·s
Rel. precision	**0.11%**

Implied Planck's constant:

e used	Implied h
Millikan's e (1916)	6.573×10^{-34}
Modern NIST e	6.614×10^{-34}
Modern exact	$\mathbf{6.626 \times 10^{-34}}$

Result

h recovered to **0.5%** of its modern value from **5 data points**.

MLE = OLS Under Gaussian Errors

Parametric model:

$$V_{0i} = \alpha + \beta\,\nu_i + \varepsilon_i, \quad \varepsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$$

Log-likelihood:

$$\ell(\alpha, \beta, \sigma) = -\frac{n}{2}\ln(2\pi) - n\ln\sigma - \frac{1}{2\sigma^2}\underbrace{\sum_i (V_{0i} - \alpha - \beta\nu_i)^2}_{\mathrm{RSS}(\alpha,\beta)}$$

Maximising ℓ over (α, β) minimises RSS—exactly OLS.

$$\boxed{(\hat{\alpha}_{\mathrm{MLE}}, \hat{\beta}_{\mathrm{MLE}}) = (\hat{\alpha}_{\mathrm{OLS}}, \hat{\beta}_{\mathrm{OLS}})}$$

Implication

Every OLS result carries over to MLE. The MSE training loss used in neural-network regression *is* the Gaussian log-likelihood—the physics and the ML objective coincide.

The Cramér-Rao Bound and Experimental Design

Fisher information and Cramér-Rao bound for $\beta = h/e$:

$$\mathcal{I}_{\beta\beta} = \frac{S_{\nu\nu}}{\sigma^2}, \qquad \mathrm{Var}(\tilde{\beta}) \geq \frac{\sigma^2}{S_{\nu\nu}}, \quad S_{\nu\nu} = \sum_i (\nu_i - \bar{\nu})^2$$

OLS achieves this bound (Gauss-Markov + Cramér-Rao).

Dataset	n	$S_{\nu\nu}$ (10^{28} Hz^2)	SE($\hat{\beta}$) (10^{-18} V·s)
Restricted (4047–2535 Å, like Hughes)	3	1.96	10.0
Millikan 5-pt (5461–3126 Å)	5	9.28	4.6
Millikan full (5461–2535 Å)	6	28.14	2.6

Design lesson

Span frequencies widely. The CR bound justifies Millikan's choice of a nearly 2× range in ν over the narrow 60 nm UV band used by predecessors.

Universality: Zero-Shot Transfer Across Metals

The quantum hypothesis predicts: $\beta = h/e$ is the same for *all* metals. This is a strong, falsifiable structural restriction.

Separate regressions:

$$\hat{\beta}_{\mathrm{Na}} = 4.129 \times 10^{-15}\ \mathrm{V{\cdot}s}$$
$$\hat{\beta}_{\mathrm{Li}} = 4.137 \times 10^{-15}\ \mathrm{V{\cdot}s}$$
$$\text{difference} = 0.19\%$$

Wald test of $H_0 : \beta_{\mathrm{Na}} = \beta_{\mathrm{Li}}$:

$$z \approx 0.87 \quad (\ll 1.96)$$

Cannot reject universality.

ML Language

Pre-training: fit slope on sodium

Zero-shot transfer: apply slope to lithium (different intercept only)

Fine-tuning needed: only intercept α_{Li} (work function shifts)

Transfer error: 0.2%

The pooled estimate $\hat{\beta}_{\mathrm{pool}} \approx 4.133 \times 10^{-15}\,\mathrm{V{\cdot}s}$ has SE reduced by $\approx 1/\sqrt{2}$ relative to either single-metal estimate.

Historical Estimates of Planck's Constant

Authors	Year	h (10^{-27} erg·s)	Error
Hughes	1912	3.55–5.85	−12 to −46%
Richardson & Compton	1912	3.5–5.8	−12 to −47%
Planck (black body)	1913	6.415	−3.2%
Millikan (Na, 5-pt)	1916	6.573	−0.80%
Millikan (Na, 9-pt)	1916	6.578	−0.73%
Millikan (Li)	1916	6.584	−0.64%
Millikan mean	**1916**	**6.579**	**−0.72%**
Modern NIST 2022	2022	6.6261	0

Pattern: pre-Millikan estimates biased downward 10–47%.

Three causes: stray UV (inflates V_0 at long wavelengths $\Rightarrow$ flattens slope), narrow $\Delta\nu$ (large SE), uncontrolled v_c (slope bias).

Millikan solved all three.

Millikan's Epistemic Reluctance

Millikan confirmed Einstein's equation to 0.5%.
Then wrote:

> *"Despite then the apparently complete success of the Einstein equation, the physical theory of which it was designed to be the symbolic expression is found* **so untenable** *that Einstein himself, I believe, no longer holds to it."*
>
> — Millikan (1916), p. 388

He had hoped the experiment would **refute** the equation!

Lesson for Statistics

Confirming a regression equation is **not** the same as confirming the structural model that generated it.

Millikan confirmed the slope h/e to 0.5% without accepting that light consists of quanta.

Compton scattering (1923) finally convinced him.

Nobel acceptance (1923):
"...contrary to my own expectation, **first direct experimental proof** ...of the complete validity of the Einstein equation."

The ML-to-Physics Dictionary

Scientific step	**ML name**	**Who**
Propose $E = h\nu$; derive $V_0 = (h/e)\nu - W/e$	Inductive bias + Hypothesis class	Einstein (1905)
V_0 depends on ν not I	Feature engineering	Lenard (1902)
Measure (ν_i, V_{0i}) pairs	Labeled training data	Millikan (1916)
Control contact EMF v_c	Nuisance-variable identification	Millikan (1916)
Min. $\sum(V_{0i} - \hat{V}_{0i})^2$	MSE training (OLS/MLE)	Millikan; this paper
$\lvert\hat{\beta} - h/e\rvert < 1\%$	Test accuracy / Confirmation	Millikan (1916)
Na and Li give same $\hat{\beta}$	Zero-shot transfer learning	Millikan (1916)

The Hypothesis Class: Where Einstein Differs from a Naive ML System

Einstein's hypothesis class:

$$\mathcal{H} = \left\{ f_{\alpha,\beta} : \nu \mapsto \alpha + \beta\nu \;\middle|\; \alpha \in \mathbb{R},\ \beta > 0 \right\}$$

with the *cross-metal restriction* that β is universal.

- VC dimension $= 2$
- One-parameter family of parallel lines
- Derived *deductively* from $E = h\nu$
- Five training points are *more than sufficient*

What a Naive ML System Misses

A linear regression package gives the right numerical answer—but:

1. Cannot identify $\hat{\beta} = 4.129 \times 10^{-15}$ as h/e
2. Cannot impose $\beta_{\mathrm{Na}} = \beta_{\mathrm{Li}}$ *from first principles*
3. Cannot decompose intercept into W/e vs. v_c
4. Cannot reduce data requirement from $O(10^3)$ to 5

The Reduction Einstein Achieved

$E = h\nu$ reduced the hypothesis space from the infinite-dimensional function class to a **two-dimensional manifold of parallel lines**— compressing the data requirement by three orders of magnitude.

Einstein vs. a Deep Neural Network: Three Epochs

Epoch	n avail.	Einstein model	DNN (VC $\approx$ 5700)	DNN outcome
Lenard 1902	≈ 0	Sufficient†	Insufficient	Cannot train (no labels)
Hughes 1912	~ 30	Sufficient	Insufficient ($\times 190$)	Memorises; no generalisation
Millikan 1916	5	Sufficient	Catastrophically insuf.	Interpolates; no inference
Modern era	~ 5000	Sufficient	Sufficient	Recovers line; cannot identify h/e

†Functional form derived from theory; only two free parameters remain.

A three-layer DNN (16 neurons/layer) has $p \approx 593$ parameters and VC dim ≈ 5700. Reliable 1% generalisation requires $n \gtrsim 5700$—short by three orders of magnitude at Millikan's epoch.

The DNN on Millikan's Five Points

What happens if you train any DNN on $n = 5$ data points?

Interpolation regime (any DNN with $p > 5$):

- Training loss $= 0$ (memorises all points)
- Function is unconstrained *between* points
- Cannot estimate slope uncertainty: zero residual df

OLS with $n = 5$, $p = 2$:

- 3 residual degrees of freedom
- $\mathrm{SE}(\hat{\beta}) = 4.6 \times 10^{-18}$ V·s
- 0.11% relative precision, at the Cramér-Rao bound

The Core Lesson

Einstein's theory (not more data, not a better algorithm) is what made five training points sufficient.

Specifying the hypothesis class was a harder intellectual act than fitting the line.

Where DNNs Win

High-dimensional problems where no parsimonious parametric family is known (images, proteins, language).

A Broader Lesson: The Hypothetico-Deductive Loop

Step	Science	ML
Propose $E = h\nu$	Theoretical hypothesis	Model architecture
Derive $V_0 = (h/e)\nu - W/e$	Deduction	Hypothesis class
Choose ν, not I	Experiment design	Feature engineering
Measure (ν_i, V_{0i})	Observation	Training data
Minimise RSS	Least squares	MSE training
$\lvert\hat{\beta} - h/e\rvert < 1\%$	Confirmation	Test accuracy
Na and Li: same $\hat{\beta}$	Universality test	Zero-shot transfer

Modern AI for Science — the Missing Piece

Modern systems (AI Feynman, symbolic regression) can *induce* the linear law from data by exhaustive search, but they do so over a combinatorially large function space. Einstein's deductive step encoded the result of that search in one paragraph.

The human contribution—theory construction—remains indispensable.

Bonus: Born's Rule and an Unintended Consequence

Einstein's quantum hypothesis required:

$$|E|^2 \propto \text{photon } \textit{density}$$

The classical wave intensity $|E|^2$ becomes a **probability density**, not just an energy density.

Einstein called this the *Gespensterfeld* (ghost field)—a wave that guides probabilities.

Max Born (1926) formalised the same logic for the Schrödinger wave function:

$$\boxed{|\psi|^2 = \text{probability density}}$$

Born's rule — now the cornerstone of quantum mechanics.

Two Nobel Prizes

1921: Einstein — for the equation that made the measurement possible

1954: Born — for the probabilistic interpretation the equation made *inevitable*

Einstein spent the rest of his life rejecting Born's rule as a "complete description of reality."

He won the Nobel for a paper whose logic seeded the interpretation he most opposed.

Summary and Conclusions

The Thesis, Revisited

Einstein was doing machine learning in 1905—but the most important step was the one no ML system has yet automated: **specifying the hypothesis class**.

Three principles illustrated:

1. **Theoretical priors guide design.** Once $E = h\nu$ fixes the functional form, only 5 training points are needed for 0.1% precision.
2. **Data quality is the binding constraint.** The regression is trivial; cleaning the data took 29 years.
3. **Confirming an equation $\neq$ endorsing the model.** Millikan confirmed the slope to 0.5% while publically rejecting the light-quantum interpretation.

The Limit of Pure AI

A data-driven system given only (ν_i, V_{0i}) pairs can recover $\hat{\beta} = 4.129 \times 10^{-15}$ V·s but cannot know it is h/e. **Physical meaning requires physical theory.**

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